Sanaul Haque

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Assignment 202

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Research Topic: FISH-PFAS: Integrated Miniaturized-QuEChERS Preparation and Fluorescent Biosensor for PFAS Detection in Freshwater Fish

Title: FISH-PFAS: Integrated Miniaturized-QuEChERS Preparation and Fluorescent Biosensor for PFAS Detection in Freshwater Fish

Research Problem

Perfluoroalkyl and Polyfluoroalkyl Substances, or PFAS, are a large group of indestructible, toxic chemicals made of strong C-F bonds that are commonly used in non-stick cookware and waterproof swimsuits. As a large network of industrial discharges from these products, the pollution of PFAS becomes a huge issue, contaminating surrounding ecosystems and water pathways. In this case, the pollution of freshwater fish remains one of the greatest pressing problems, which are potentially caught and consumed, leading to bigger public health risks. With a notable increase in fish sectors, many regions across the surface heavily rely on local fishery, one of which is North Sulawesi. As the central hub for maritime activities, fisheries sustain both the local economy and daily dietary protein. According to the Marine and Fisheries Department of North Sulawesi [1], the annual fish consumption per person reaches 48.97 kilograms, more than double the global average of only 20.7 kilograms per person each year as reported by the Food and Agricultural Organization [2]. With emerging industrial runoff introducing PFAS into the local sea waters, the extreme reliance on local seafood creates a significant vulnerability in public health.

While many solutions have been proposed in the detection of PFAS, the standard detection remains extremely slow for market testing, as fast detection is needed to monitor the existence of PFAS in captured freshwater fish right before market distribution. A considerable gap also exists between the matrix gap, where high lipid and protein levels in fish tissues potentially cause signal interference, obstructing the sampling method procedure required for a precise detection. Thus, this research aims to create an ideal tool for PFAS detection in freshwater fish by integrating a miniaturized-QuEChERS sample preparation with a handheld fluorescent biosensor to strip away lipid and protein interference and enable a rapid and accurate screening.

Existing Literature

Previous studies have presented a clear operational mechanism of PFAS detection in freshwater samples. In 2021, Mann et al. [3] managed to engineer a protein-based optical biosensors from human liver fatty acid-binding protein (cpGFP-hLFABP). This allows the protein to bind and transport fatty acids within cells, which works for the binding of Perfluorooctanoic Acid (PFOA), a type of PFAS that shares similar cell characteristics as fatty acids, thus enabling rapid and optical detection of PFAS in freshwater samples. This

method was studied and developed once more by Mann and Berger in 2023 [4], before deployed directly in measuring the levels of PFAS in 2024 by Mann et al. [5]. While recent developments and records have shown a promising nature of this device, Sankhla et al. [6] have reiterated that the gold-standard regulatory of PFAS detection and monitoring relies heavily on Liquid-chromatography Tandem Mass Spectrometry (LC-MS/MS), which requires costly standard equipment and laboratory logistics.

On the other hand, many studies have also developed sample preparation methods for freshwater fish. Hassan et al. [7] have demonstrated the use of the QuEChERS (Quick, Easy, Cheap, Effective, Rugged, Safe) method, which remains the most popular and efficient extraction technique. Koloka et al. [8] further used this technique to analyze PFAS residue in fish and fish feed, which shows an optimized sample preparation. This shows the primary inquiry tests whether a miniaturized-QuEChERS cleanup can selectively strip away background lipids without compromising biosensor readouts in fish tissue. Thus, this study fills both literature gaps by establishing the analytical approach by benchmarking fluorescent cpGFP-hLFABP biosensor signals directly against LC-MS/MS data to evaluate recovery efficiency and signal linearity using quantitative approaches.

Research Question

How does integrated miniaturized-QuEChERS preparation combined with a cpGFP-hLFABP fluorescent biosensor determine PFAS concentration in local freshwater fish compared to standard LC-MS/MS analysis?

Methodology

This study adopts a quantitative experimental research approach to ensure a thorough understanding of the analytical performance and detection capability:

Sampling Methods

- Methods: This research uses stratified random sampling across major public fish markets and landing sites in North Sulawesi.
- Time: Fish samples used will be collected multiple times of day (morning and afternoon batches) and varying days of the week.
- Specimen Variety: All fish samples will be selected across diverse species, sizes, and weight classes to reflect real-world bioaccumulation.
- Vendor Distribution: Samples will be purchased randomly across independent, local vendors rather than a single supplier.

Methodology and Experimental Framework:

- Controlled laboratory research is conducted by manipulating the PFAS spike concentrations across the locally collected fish samples while evaluating optical fluorescence signals and recovery rates.
- Empirical precision is established by conducting signal-to-noise ratios and limits of detection to prove system reliability before field deployment.
- Experimental framework of this research is done by in-vitro tissue validation, which directly benchmarks the biosensor readouts against gold-standard LC-MS/MS data to ensure accuracy and precision.

Ethical Considerations:

- Data and privacy of fish merchants will be kept confidential.
- All fish samples used are purchased standard over the counter to avoid illegal harvesting.
- All findings of the research will be routed through public health agencies to provide actional consumption guidance.
- Testing focuses solely on market-procured fish tissue, completely avoiding any invasive procedures that involve biological testing.

Research Topic

This topic is immediately pressing at the regional level. With the overwhelmingly high number of fish consumption rising each year in North Sulawesi, the need to develop a fast and reliable detection tool for PFAS monitoring in locally harvested freshwater fish is in great demand. By constructing an integrated miniaturized- QuEChERS method with cpGFP-hLFABP fluorescent biosensor, this research offers an innovative component in validating a novel optical PFAS sensor. This technology could give coastal communities a more accessible way to monitor PFAS contamination, and rather than relying entirely on distant laboratories, communities could potentially conduct preliminary testing themselves and identify changes or areas of concern. This research potentially could support community knowledge, inform safer fish-consumption decisions, and provide locally generated evidence to guide further environmental and public health monitoring.

Quality of Writing

This proposal is written with academic precision and clear expression, appropriate for a multidisciplinary audience. Jargon is minimized, and the structure adheres by standard

research normals. The proposal complies with the IARCO standards, thus aligning with the institutional's grammar, citation, and formatting guidelines as strictly as possible.

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